
id: MDCS-TICKET-005
title: Create Python idioms reference drawers.
epic: epic-dumb-copilot
status: ready
risk: low
allowed_areas:
- references/python-idioms/
must_not_touch:
- tools/
- procedures/
requirements:
- Create the Python idioms reference directory structure: **references/python-idioms/sections/**.
- Create **references/python-idioms/README.md** as the index. It must:
* Declares the reference scope and consuming skills.
* Limit the index size to at most 120 lines total.
* Contain a markdown Sections table with columns: **id | topic | open when** (note the exact space in “open when”).
* Map the following drawers: **pandas_etl** and **openpyxl_excel**.
- Create **references/python-idioms/sections/pandas_etl.md**:
* Limit size to at most 150 lines total.
* Begin with a one-line **relevant-when:** header key-style line.
* Focus on connecting python to Oracle using **oracledb** in Thin mode.
* Provide patterns for executing select queries, cursor fetching, and mapping to pandas DataFrames.
* Handle memory allocation (e.g., chunking, using generators) and type conversion safety (e.g., converting decimals, mapping On
- Create **references/python-idioms/sections/openpyxl_excel.md**:
* Limit size to at most 150 lines total.
* Begin with a one-line **relevant-when:** header key-style line.
* Provide patterns to format Excel workbooks generated via **openpyxl** (fonts, borders, cell fills, number formats).
* Show how to auto-fit columns, freeze panes, apply filters, and write multi-sheet workbooks.
non_goals:
- Do not cover generic python algorithms or frontend libraries. Focus strictly on Oracle-Pandas ETL and openpyxl formatting.
acceptance_criteria:
- Running **python tools/validate_mdcs_library.py** passes on the Python references.
- The index is 120 lines and sections are 150 lines.
- Every drawer begins with a one-line **relevant-when:** header.
verification_commands:
- python tools/validate_mdcs_library.py
depends_on:
- MDCS-TICKET-001

Body

Goal

Compile standard, dense Python reference drawers to feed Copilot during active procedures for ETL and reporting tasks.

Context

Writing data-loading and Excel-formatting scripts is a repetitive IT-BA task. Edge Copilot often produces outdated **cx_Oracle** code or unstyled Excel sheets. Providing these modern drawers ensures Copilot writes robust, copy-pasteable Python code.

Procedure

1. Initialize **references/python-idioms/README.md** and write the index table.

2. Draft `references/python-idioms/sections/pandas_etl.md` starting with a `relevant-when:` header and showing how to execute database reads and convert cursors to dataframes using the modern `oracledb` thin mode.
3. Draft `references/python-idioms/sections/openpyxl_excel.md` starting with a `relevant-when:` header and detailing professional spreadsheet styling.
4. Run the validator tool to verify formatting and line count constraints.